SkillFolio — RunPod Proof of Use

Team MIX MECHS · SDC Hackathon 2026 · evidence collected 2026-10-24

1. What runs on RunPod

Item	Value
Service	RunPod Serverless endpoint `r0ykxnttw93erm` (name: `skillfolio`)
Hardware	16 GB GPU (RTX A4000 class), 0 always-on workers, max 1 worker (scales to zero when idle)
Model	`BAAI/bge-small-en-v1.5` (open embedding model, MIT licence), baked into the Docker image
Job	Resume-to-skills matching: each resume sentence is compared with a plain description of each catalogue skill; only catalogue skill IDs can come back
Called from	The app (app/runpod_client.py) when the two RunPod keys are set. The public Replit demo runs without them, in keyword mode. Falls back to keywords after an error or 40 s.

2. Live calls to the endpoint

Two requests sent on 2026-10-24 with exactly the payload the app sends. The resume never names a tool.

Call	Started (UTC)	Status	Queue / start-up	GPU execution	Total	Skills returned (similarity)
1 (cold worker)	2026-10-24 05:20:18 UTC	COMPLETED	13.8 s	0.54 s	15.1 s	sql (0.81), powerbi (0.78), excel (0.75)
2 (warm worker)	2026-10-24 05:20:48 UTC	COMPLETED	1.6 s	0.29 s	2.5 s	sql (0.81), powerbi (0.78), excel (0.75)

Endpoint health after the calls (RunPod `/health`): **16 jobs completed, 0 failed**; workers ready: 1.

Resume text sent: *Business analytics graduate. Built dashboards for senior management to track monthly sales. Cleaned and merged spreadsheets from three branches. Wrote queries to pull customer data from the company database.*

"Built dashboards for senior management" → Power BI; "cleaned and merged spreadsheets" → Excel; "wrote queries … database" → SQL.

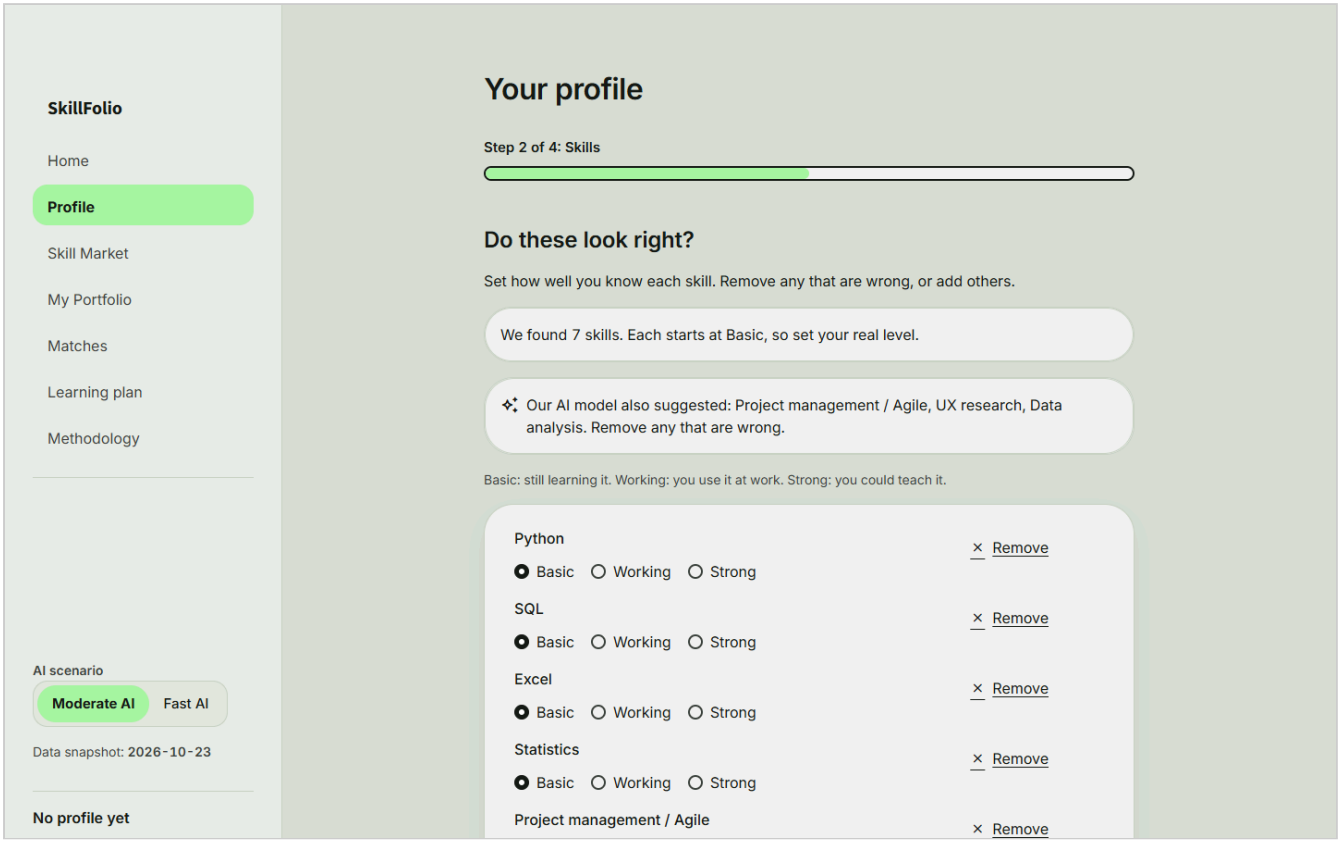
Request (API key omitted)

```
{
 "input": {
  "task": "match_skills",
  "text": "Business analytics graduate. Built dashboards for senior management to track monthly sales.
Cleaned and merged spreadsheets from three branches. Wrote queries to pull customer data from the company
database.",
  "skills": "[73 catalogue skills, each with one plain sentence, e.g. {'skill_id': 'powerbi', 'text':
'Building dashboards and reports for managers in Microsoft Power BI.'}]",
  "threshold": 0.73,
  "top_k": 6
 }
}
```

Response from RunPod (call 1)


```
{
 "id": "sync-bb81878f-ade3-4c2d-baaf-b89cce01b406-e2",
 "status": "COMPLETED",
 "delayTime": 13824,
 "executionTime": 543,
 "workerId": "isxfj9khicn0g3",
 "output": {
  "matches": [
   {
    "score": 0.811,
    "skill_id": "sql"
   },
   {
    "score": 0.78,
    "skill_id": "powerbi"
   },
   {
    "score": 0.751,
    "skill_id": "excel"
   }
  ],
  "model": "BAAI/bge-small-en-v1.5"
 }
}
```


3. In the app



Resume step: skills found by keyword, then the RunPod model's extra suggestions, which the user can remove.

4. Deployment code

`runpod/handler.py`:

```
"""RunPod serverless handler: resume-to-skills matching with an open embedding model on a GPU.

Deployed as a RunPod Serverless Endpoint from runpod/Dockerfile (see runpod/README.md). The logic lives in
skill_match.py.
"""

import runpod
import torch
from sentence_transformers import SentenceTransformer

from skill_match import MODEL_NAME, run

_model = SentenceTransformer(MODEL_NAME, device="cuda" if torch.cuda.is_available() else "cpu")   # loaded
once per worker


def embed(texts):
    return _model.encode(list(texts), normalize_embeddings=True, batch_size=64)


def handler(job: dict) -> dict:
    return run(job, embed)


runpod.serverless.start({"handler": handler})
```

runpod/Dockerfile:

```
# RunPod Serverless worker for SkillFolio: resume-to-skills matching on a GPU.
# Build context = the repo root (RunPod console: Dockerfile path "runpod/Dockerfile", build context ".").
FROM python:3.11-slim

ENV PYTHONUNBUFFERED=1 HF_HOME=/models
WORKDIR /worker

COPY runpod/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Bake the model into the image so a cold worker does not download it (faster start, no network needed).
RUN python -c "from sentence_transformers import SentenceTransformer; SentenceTransformer('BAAI/bge-small-
en-v1.5')"

COPY runpod/skill_match.py runpod/handler.py ./
CMD ["python", "-u", "handler.py"]
```

Source: https://github.com/SDC-HACKATHON-MIX-MECHS/skillfolio (runpod/, app/runpod_client.py). No API key appears in this document or the repository.